

In the **AI context**, a *transformer* is a deep learning architecture that revolutionized natural language processing (NLP) and beyond. Instead of processing data sequentially like older models (RNNs, LSTMs), transformers rely on **attention mechanisms** to capture relationships between all elements in a sequence at once.

Core Ideas Behind Transformers

Input Representation: Words (or tokens) are converted into vectors using embeddings.

Self-Attention: Each token looks at every other token in the sequence to decide which ones are most relevant.

Multi-Head Attention: Multiple "attention heads" learn different types of relationships (syntax, semantics, context).

Positional Encoding: Since transformers don't process sequentially, they add position information so the model knows word order.

Feedforward Layers: After attention, data passes through fully connected layers for transformation.

Stacked Layers: Many layers of attention + feed forward build deep contextual understanding.

How It Works Step by Step

Tokenization: Break text into tokens.

Embedding + Position Encoding: Convert tokens into vectors with positional info.

Self-Attention: Compute relevance scores between tokens.

Example: In "*The cat sat on the mat*", the word "*cat*" pays attention to "*sat*" and "*mat*" more than "*the*".

Weighted Combination: Each token becomes a mix of other tokens based on attention weights.

Feedforward Network: Adds non-linear transformations.

Stack Layers: Repeat attention + feedforward many times.

Output Layer: Produces predictions (next word, classification, translation, etc.).

⚡ Why Transformers Are Powerful

Parallelization: Unlike RNNs, transformers process all tokens simultaneously → faster training.

Long-Range Dependencies: Attention lets the model connect distant words (e.g., subject–verb agreement across long sentences).

Scalability: Forms the basis of large language models (like GPT, BERT, etc.).

Applications

Language Models: Chatbots, translation, summarization.

Vision Transformers (ViT): Image recognition.

Multimodal Models: Combine text, images, audio.

Generative AI: Text generation, code completion, creative writing.

In short: **Transformers replaced sequential memory with attention, enabling models to "see the whole picture" at once.** That's why they became the backbone of modern AI.

a) Self-Attention Mechanism

Purpose: Helps the model decide which words in a sentence are most relevant to each other.

How it works:

- Each token is transformed into three vectors: **Query (Q)**, **Key (K)**, and **Value (V)**.
- The model calculates similarity between Q and K to get *attention scores*.
- These scores weight the V vectors, producing a new representation of each token that incorporates context.

Example: In “*The cat sat on the mat*”, the word “*cat*” attends strongly to “*sat*” and “*mat*”, less to “*the*”.

b) Vector Embedding

Purpose: Converts discrete tokens (words, subwords, characters) into continuous numerical vectors.

How it works:

- Each token is mapped to a high-dimensional vector space.
- Similar words (like *king* and *queen*) end up with embeddings close together.
- Transformers add **positional encoding** so the model knows word order.

Benefit: Embeddings capture semantic meaning, allowing the model to understand relationships beyond exact word matches.

c) Feed Forward Neural Network (FFNN)

Purpose: Adds non-linear transformations after attention to refine representations.

How it works:

- Each token's vector passes through a small fully connected neural network (same network applied independently to each token).
- Typically: Linear → Activation (like ReLU or GELU) → Linear.
- **Benefit:** Increases the model's expressive power, enabling it to capture complex patterns.

Putting It All Together

- **Embeddings** turn words into vectors.
- **Self-attention** mixes information across tokens, highlighting relevant context.
- **Feedforward layers** refine these contextualized vectors.

Stacking many layers of these components builds deep understanding of language.